CH 3 — Diagonalizability

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Warm Up Question

Let $R : \mathbb{R}^2 \to \mathbb{R}^2$ be reflection in the line $y = 3x$. Determine $R(\vec{v})$

Reflection is a linear map. R can be represented by a 2×2 matrix.

Choose a basis $B = \{e_1, e_2\} \in \mathbb{R}^2$. Build ${}_B[A]_B$

$A = [R]_B = \begin{bmatrix} [R(e_1)]_B & [R(e_2)]_B \end{bmatrix}$. But we don't know $R(e_1), R(e_2)$.

We choose instead the vector $R(v_1) = v_1$, that is on the line $y = 3x$, say $v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

Then choose another vector $R(v_2) = -v_2$, that is perpendicular to the line $y = 3x$, say $v_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$

$D = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}$

Now $C = {}_D[R]_D = \begin{bmatrix} [R(v_1)]_D & [R(v_2)]_D \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Lesson:

1. Sometimes there may be a "nice" basis for $L : V \to V$ that makes computations easier
2. Dream scenario: want the basis D so that we have $[L]_D = \begin{bmatrix} x & \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots & x \end{bmatrix}$

> 📘 **Definition — Operator**
>
> Let $T : V \to W$ is called a **linear operator** if $\dim V = \dim W$

> 📘 **Definition — Diagonalizability**
>
> Compute the matrix of $L : V \to V$ and obtain the basis D so that we have $[L]_D = \begin{bmatrix} x & \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots & x \end{bmatrix}$
>
> If there is such a basis D, we say L is **diagonalizable** and we say D is a **diagonalizable basis**

Example:

Reflection map $R : \mathbb{R}^2 \to \mathbb{R}^2$ across $y = 3x$ is diagonalizable and $D = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}$ is a diagonalizable basis.

Suppose we want to calcualte $R(\vec{v})$.

$[R(\vec{v})]_D = [R]_D [\vec{v}]_D$

But we want $R(\vec{v}) = [R(\vec{v})]_S, S = \{e_1, e_2\}$

Let ${}_S I_D$ be change for D to S coordinates. Then ${}_S I_D [\vec{x}]_D = [\vec{x}]_S$

We obtain

$$
\begin{aligned}
{}_S I_D[R(\vec{v})]_D &= {}_S I_D[R]_D[\vec{v}]_D \\
[R(\vec{v})]_S &= {}_S I_D[R]_D[\vec{v}]_D \\
[R(\vec{v})]_S &= {}_S I_D[R]_D({}_S I_D)^{-1}{}_S I_D[\vec{v}]_D \\
[R(\vec{v})]_S &= {}_S I_D[R]_D({}_S I_D)^{-1}[\vec{v}]_S
\end{aligned}
$$

So

$$
{}_S[R]_S = {}_S I_D{}_D[R]_D{}_D I_S
$$

${}_S I_D = \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$

$\therefore [R]_S = \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\left(\begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}\right)^{-1} = \frac{1}{10}\begin{bmatrix} -8 & 6 \\ 6 & 8 \end{bmatrix}$

📘 **Definition — Similarity**

Let $A, B \in M_{n\times n}(\mathbb{F})$, we say A is **similar** to B if $\exists$ invertible matrix P s.t. $A = PBP^{-1}$

If $A \sim B$

1. rank A = rank B
2. nullity A = nullity B
3. det A = det B
4. tr A = tr B

Example:

If B, C are bases for V and if $L : V \to V$ is linear, then ${}_B[L]_B \sim {}_C[L]_C$

since ${}_B[L]_B = {}_B I_C{}_C[L]_C({}_B I_C)^{-1}, P = {}_B I_C$

QUESTION: Is $[L]_B$ for any choice of B, similar to a diagonal matrix?

Warm Up

Suppose $D = v_1, ..., v_n$ is a diagonalizable basis for L. That is suppose $[L]_D = \begin{bmatrix} \lambda_1 & ... & 0 \\ ... & ... & ... \\ 0 & ... & \lambda_n \end{bmatrix}$

$\iff \left[[L(v_1)]_D ... [L(v_n)]_D\right] = \begin{bmatrix} \lambda_1 & 0 & ... & 0 \\ 0 & \lambda_2 & ... & 0 \\ ... & ... & ... & ... \\ 0 & ... & 0 & \lambda_n \end{bmatrix}$

Equating cols, $[L(v_1)] = \begin{bmatrix} \lambda_1 \\ 0 \\ ... \\ 0 \end{bmatrix} \iff L(v_1) = \lambda_1 v_1 + 0v_2 + ... + 0_n, \therefore L(v_1) = \lambda_1 v_1$

Similarly, $L(v_1) = \lambda_i v_i, i = 1, 2, ..., n$

So: $D = \{v_1, ..., v_n\}$ is a diagonalizable basis for L if and only $L(v_1) = \lambda_i v_i, \forall i$ if and only v_i is an eigenvector for L with eigenvalue λ

📘 **Definition — Eigenbasis**

An **eigenbasis** for L is a basis $\{v_1, ..., v_n\}$ where the v_i are eigenvectors.

(In this case $[L]_D = \begin{bmatrix} \lambda_1 & ... & 0 \\ ... & ... & ... \\ 0 & ... & \lambda_n \end{bmatrix}$ and the i-th diagonal entry λ_i is the eigenvalue of v_i)

Examples:

1. $R : \mathbb{R}^2 \to \mathbb{R}^2$ reflection across $y = 3x$

   We found $v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is on the line and $R(v_1) = v_1$, so v_1 is an eigenvector with $\lambda_1 = 1$

   Also, $v_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ is perpendicular to the line and $R(v_2) = -v_2$ is an eigenvector with $\lambda_2 = -1$

   Therefore, $D = \{v_1, v_2\}$ is an eigenbasis. We computed $[L]_D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ as expected.

2. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation by $\theta \neq \pi, 2\pi, 0 < \theta < 2\pi$

   Then T has no eigenvectors so T has no eigenbasis and T is not diagonalizable

3. Let $D : P_2(\mathbb{R}) \to P_2(\mathbb{R}), D(p(x)) = p'(x)$

   Does D have any Eigenvectors

   Yes, if $p(x) = c, D(p(x)) = 0 = 0 \cdot p(x)$ is an eigenvector with $\lambda = 0$

   If $\deg p > 1$, then $D(p) \neq \lambda p, \lambda \neq 0$.

   The only possible eigenvalue is $\lambda = 0$ when eigenvector are constant.

   $\therefore D$ is not diagonalizable.

   Note, if D : functions $\to$ functions, $D\left(e^{\lambda x}\right) = \lambda e^{\lambda x}$

4. Let $D : P_2(\mathbb{R}) \to P_2(\mathbb{R}), D(p(x)) = p'(x)$

   We work with matrix, $S = \{1, x, x^2\}$

   $[D]_S = \left[[D(1)]_S \ [D(x)]_S \ [D(x^2)]_S \right] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

   What is the relationship between eigenvectors and eigenvalues of D and those of $A = [D]_S$?

   $D(p) = \lambda p \implies [D(p)]_S = [\lambda p]_S \iff [D]_S[p]_S = \lambda[p]_S \iff A[p]_S = \lambda[p]_S$

   $\therefore D(p) = \lambda p \iff A[p]_S = \lambda[p]_S$

📐 **Theorem —**

Let $L : V \to V$ be an operator. Let B be a basis for V, $A = [L]_B$

Then $v \in V$ is an eigenvector for L with eigenvalue $\lambda \iff [v]_B$ is an eigenvector for A with eigenvalue λ

💡 **Tip** — Linear Operators

Whenever struggling to solve a question with linear operators, translate into matrix language and attempt to solve it.

Example:

$D : P_2(\mathbb{R}) \to P_2(\mathbb{R}), A = [D]_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$.

Notice this is a triangular matrix, the eigenvalue are the diagonal entries.

$\lambda = 0$ is the only eigenvalue for A

Therefore $\lambda = 0$ is the only eigenvalue for D

$Av = \lambda v \iff Av - \lambda v = 0 \iff (A - \lambda I)v = 0 \iff v \in \text{Null}(A = \lambda I)$

Eigenvectors $\begin{bmatrix} c \\ 0 \\ 0 \end{bmatrix}$

Definition — Eigenspace

The **eigenspace** of L with eigenvalue λ is

$$E_\lambda(L) = \{v \in V : L(v) = \lambda v\}$$

for all λ-eigenvectors and $\vec{0}$

The eigenspace of $A \in M(n \times n)(\mathbb{F})$ with eigenvalue λ is

$$E_\lambda(A) = \{x \in \mathbb{F}^n : Ax = \lambda x\}$$

$E_\lambda(L)$ is subspace of V

$E_\lambda(A)$ is a subspace of $\mathbb{F}^n$

Theorem — Isomorphic Eigenspaces

Proposition

Suppose $L : V \to V$ is linear and let $A = [L]_B$ then $E_{\lambda(L)} \cong E_{\lambda(A)}$

Corrollary

1. $\dim E_\lambda(L) = \dim E_\lambda(A)$ is the geometric multiplicity of λ
2. $B = \{v_1, ..., v_n\}$ is a basis for $E_\lambda(L) \iff \left\{[v_1]_B, ..., [v_n]_B\right\}$ is a basis for $E_{\lambda(A)}$

Proof

Define $[] : E_{\lambda(L)} \to E_{\lambda(A)}, v \mapsto [v]_B$. This has an inverse by untaking coordinates. So this map is an isomorphism

□

Let $A \in M_{n \times n}(\mathbb{F})$ That is we want $P = [v_1 \ ... \ v_n]$ and $D = \begin{bmatrix} \lambda_1 & ... & 0 \\ ... & ... & ... \\ o & ... & \lambda_n \end{bmatrix}$ so that $A = PDP^{-1}$.

v_i, λ_i are eigenvectors and eigenvalues. We need eigenvalues to build D and linear independent eigenvectors for P

Steps

1. Calculaet and factor the characteristic polynomial of A

   $C_A(\lambda) = \det(A - \lambda I) = ...$ If cannot factor into linear terms. Then A is not diagonalizable. STOP.

   The roots of $C_A(\lambda)$ are eigenvalues: $\{\lambda_1, \lambda_1, ..., \lambda_1\}, ..., \{\lambda_n, \lambda_n, ..., \lambda_n\}$. The number of times each distinct eigenvalues appear is the algebraic multiplicity.

   Example: $C_\lambda(A) = \det(A - \lambda I) = (\lambda)^3(\lambda - 1)^2(\lambda - 5)$. Thus $a_0 = 3, a_1 = 2, a_2 = 1$

   Then we obtain $D = \begin{bmatrix} \lambda_1 & ... & 0 & 0 \\ 0 & \lambda_1 & ... & 0 \\ ... & ... & ... & ... \\ 0 & ... & 0 & \lambda_n \end{bmatrix}$

2. Determine the eigenspaces. $E_{\lambda(A)} = \{x : (A - \lambda I)x = 0\} = \text{Null}(A - \lambda I)$

   If $g_\lambda \neq a_\lambda$, STOP, A is not diagonalizable

   If $g_\lambda = a_\lambda, \forall \lambda$, A is diagonalizable

   Find bases for each eigenspaces and put them down as columns in P

Example:

1. $D : P_2 \to P_2$ is differentiation

   If $S = \{1, x, x^2\}$, then $[D]_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \lambda = 0, 0, 0$

   We also saw that $E_0(D) = \text{Null}(D - 0I) - \text{Null}(D) = \text{span}\{1\}, g_\lambda = 1 \neq a_\lambda = 3$

   D not diagonalizable.

2. $L : P_2(\mathbb{R}) \to P_2(\mathbb{R}), L(p(x)) \mapsto p(1)(1 + x)$

   $S = \{1, x, x^2\}, A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

   $C_A(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{bmatrix} = (\lambda - 3)(\lambda - 1)^2, \lambda = 1, 1, 3 \implies a_1 = 2, a_3 = 1$

   $\lambda = 1$, find basis for $E_\lambda(A) = \text{Null}(A - \lambda I) = \text{Null}(A - I) = \text{Null}\left(\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \right)$

   $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\} \implies g_1 = 2 = a_2; Av = \lambda v \implies \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

   $\lambda = 3, E_\lambda(A) = \text{Null}(A - 3I) = \text{Null}\left(\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \right) \implies \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\} \implies g_3 = 1 = a_1$

   $L : P_2(\mathbb{R}) \to P_2(\mathbb{R})$: untake S −coordinates, we get eigenbasis $D = \{1 - x, 1 - x^2, 1 + x\}$

   $[L]_D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

💡 **Tip — Diagonalization Problems**

1. Always check calculation, whenever found a new outcome, verify it. (i.e. whenever obtain basis, check if $Av = \lambda v$) to avoid mistake
2. When $a_\lambda = 1$, then $g_\lambda = 1$. In general, $1 \leq g_\lambda \leq a_\lambda$. Except that if all eigenvalues are distinct, then L is diagonalizable.
3. If $[L]_B$ is triangular, and eigenvalues are distinct, then diagonalizable.